Complete split and non-split prime graph classification at 2-switch-degree four

Victor N. Schvöllner

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Editorial note. One complete prime-degree-four classification, combining original split and non-split theorems. Complete original mathematical paper138–2741, all auxiliary structural discussion, every active native inline graph figure and embedded bibliography retained verbatim. Original split theorem proof refers to preceding discussion; that complete discussion is included, and no new proof environment or argument is invented.

Editorial note. Lower-degree classifications, realization-graph consequences, global transitivity/regularity dichotomy and counts of derived realization graphs remain context, not additional counts.

Editorial note. Author retains responsibility for mathematics. No proof repair, new assumptions, independent mathematical review, graph enumeration/counterexample search or certification.

Allowed standard prerequisites (editorial).

- Degree-sequence realization graphs, 2-switches, active vertices, split graphs and Tyshkevich composition. Original active/primality/indecomposability/switch-degree definitions and graph-isomorphism/complement conventions retained. Cited realization-graph decomposition results remain original background.. Complete author local-counting and structural case discussion retained, including all inline figures; no graph enumeration, independent counterexample search or author proof certification.

1 Introduction

The systematic study of the graphs sharing a common degree sequence has long been a central theme in graph theory. A fundamental tool in this context

is the *2-switch*: given two vertex-disjoint edges ab and xy of a graph G with $ax, by \notin E(G)$, the 2-switch on $\{ab, xy\}$ replaces these two edges by ax and by (see [6]). Since it preserves degrees, it moves between the realizations of a fixed degree sequence d, which form the vertices of the *realization graph* $\mathcal{G}(d)$: the labeled graphs on a fixed vertex set with degree sequence d, two being adjacent whenever one is obtained from the other by a single 2-switch (see [1]). Classically, any two realizations of d are related by a finite sequence of 2-switches (see page 24 of [6]); equivalently, $\mathcal{G}(d)$ is connected.

Understanding the structure of $\mathcal{G}(d)$ is a rich area of research: Arikati and Peled proved in [1] that $\mathcal{G}(d)$ is Hamiltonian whenever d has majorization gap 1, Taylor studied in [15] constrained versions of the switching operation, and more recent work has focused on the connectivity of the subgraphs of $\mathcal{G}(d)$ induced by particular graph families (see [14]). A more granular, local description of $\mathcal{G}(d)$ is provided by the *2-switch-degree* (or simply the *degree*) of a graph G, denoted $\deg(G)$, which is the number of distinct 2-switches that can be performed on G. Since two distinct 2-switches on G produce two distinct graphs, $\deg(G)$ is precisely the degree of G as a vertex of $\mathcal{G}(d)$. This parameter was introduced in [12], where it is expressed in terms of the induced subgraphs of G of order 4.

The study of 2-switch transformations is closely related to the Tyshkevich canonical decomposition [16], which expresses every graph as a composition $G = G_n \circ \cdots \circ G_1$ of indecomposable graphs, in a way that is unique up to isomorphism when all the factors have at least one vertex. A key feature of this decomposition is that all its factors, with the possible exception of G_1, are split graphs. A graph S is *split* if its vertex set can be partitioned into a clique K and an independent set I; when such a partition is fixed we write (S, K, I) and we say that (K, I) is a *bipartition* for S. Split graphs arise naturally in several areas of graph theory and have been widely studied (see [8, 7, 17, 10]). For split graphs the 2-switch-degree admits a particularly simple description: it equals the number of induced P_4's.

A vertex of G is *active* if it takes part in some 2-switch on G. A graph whose vertices are all active is itself called *active*, and an indecomposable active graph is called *prime* (see [12]). Since every factor in the Tyshkevich decomposition of an active graph is prime, and since all these factors but at most one are split, prime split graphs are the basic building blocks of the whole theory. Their structure is further clarified by the *factor graph* $\Phi = \Phi(S, K, I)$, a loopless multigraph on the vertex set I that records the 2-switches available in S. This notion was introduced in [11], inspired by the A_4-structure of Barrus and West [4]; there it is shown that an active split graph S is prime if and only if Φ is connected.

Building on these foundations, we classify the graphs whose 2-switch-degree is small, that is, those that are least flexible under 2-switch transformations. Our starting point is a complete classification of the active graphs of degree at most 3 (Theorems 3.1 and 4.2); the prime ones among them are all split except C_4 and $2K_2$. This already determines the global shape of a realization graph from a single one of its local degrees: if X realizes d and $\deg(X) = k \leq 3$, then $\mathcal{G}(d)$ is vertex-transitive and k-regular if and only if T_{221} and $\overline{T_{221}}$ are not induced subgraphs of X (Corollary 7.3), the only exception up to complementation. The hypothesis $k \leq 3$ cannot be dropped: for every $k \geq 4$ there is a prime graph G_k of degree k carrying a 2-switch τ with $\deg(\tau(G_k)) = 2k - 2$ (Proposition 7.4), so from degree 4 on a single 2-switch may almost double the degree and regularity fails. Degree 4 is thus the first value at which the picture is no longer rigid, and it is the one we settle here: there are exactly 18 prime graphs of degree 4 (Corollary 6.2), 12 split (Theorem 5.1) and 6 non-split (Theorem 6.1). Finally, we determine the realization graphs of all 29 prime graphs of degree at most 4 (Theorem 7.7): up to isomorphism there are only 10, so a short list accounts for every realization graph arising in this range. Some of our tools are stated beyond degree 4; for instance, Theorem 2.13 applies to every graph of degree at most 5.

The rest of the article is organized as follows. Section 2 gathers the tools used throughout the paper. In Section 3 we classify the active graphs of degree at most 2, and in Section 4 those of degree 3. Sections 5 and 6 are devoted to the prime graphs of degree 4, split and non-split respectively. Section 7 studies the regularity of realization graphs having a vertex of degree $k \leq 4$ and classifies them.

2 Tools

Let G be a graph. We write $V(G)$ and $E(G)$ for its vertex set and its edge set, and $|G| = |V(G)|$ for its order. The complement of G is denoted by $\overline{G}$. We write $H \prec G$ to mean that H is an induced subgraph of G, and $G[W]$ (or $G[v_1, \ldots, v_k]$) for the subgraph of G induced by $W \subseteq V(G)$. The notation $G \approx H$ means that G and H are isomorphic. For $n \in \mathbb{N} = \{1, 2, \ldots\}$ we put $[n] = \{1, \ldots, n\}$, and $\dot\cup$ denotes disjoint union. The *neighborhood* of $v \in V(G)$ is $N_G(v) = \{x \in V(G) : vx \in E(G)\}$, and $\deg_G(v) = |N_G(v)|$ is the *degree* of v in G. When G is clear from the context we write N_v and d_v. A path is denoted by the juxtaposition of its vertices, so $uxyv$ is the P_4 with edges ux, xy, yv; for a cycle, the first vertex is repeated at the end. We

write $\omega(G)$ and $\alpha(G)$ for the clique number and the independence number of G. If $V(G) = [n]$, the *degree sequence* of G is $d = d(G) = (d_1, \ldots, d_n)$. If $d = (d_v)_{v=1}^n$ is the degree sequence of G, then $\overline{d} = (n-1-d_v)_{v=1}^n$ is the degree sequence of $\overline{G}$. For graph-theoretical concepts not defined here we refer the reader to [6].

For graphs G and H, we denote by $\mathcal{Q}_G(H)$ the set of induced subgraphs of G isomorphic to H, and we put $\mathcal{Q}_G = \mathcal{Q}_G(P_4) \,\dot\cup\, \mathcal{Q}_G(C_4) \,\dot\cup\, \mathcal{Q}_G(2K_2)$. If τ is a 2-switch on G, we write $\tau(G)$ for the resulting graph. The four vertices on which τ acts induce a member of $\mathcal{Q}_G$, and the next result counts the 2-switches on G accordingly.

Theorem 2.1 ([12])**.** *For every graph G,*

$$\deg(G) = 2|\mathcal{Q}_G(2K_2)| + 2|\mathcal{Q}_G(C_4)| + |\mathcal{Q}_G(P_4)|. \tag{1}$$

In particular, if G is a split graph, then $\deg(G) = |\mathcal{Q}_G(P_4)|$.

By (1), $\deg(P_4) = 1$ and $\deg(C_4) = \deg(2K_2) = 2$, so (1) can be rewritten as

$$\deg(G) = \sum_{H \in \mathcal{Q}_G} \deg(H). \tag{2}$$

Theorem 2.2. *If G is an active graph, then $4 \le |G| \le 4\deg(G)$.*

Proof. Since G is active, it is clear that $\mathcal{Q}_G \neq \varnothing$, as $\deg(G) \ge 1$. Thus, $|G| \ge 4$. On the other hand:

$$|G| = \left| \bigcup_{H \in \mathcal{Q}_G} V(H) \right| \le \sum_{H \in \mathcal{Q}_G} 4 \le 4 \sum_{H \in \mathcal{Q}_G} \deg(H) = 4\deg(G),$$

the last equality by (2). □

2.1 Split graphs and factor graphs

If (S, K, I) is a split graph and G is a graph disjoint from S, the *Tyshkevich composition* $(S, K, I) \circ G$ is the graph with vertex set $V(S) \cup V(G)$ and edge set $E(S) \cup E(G) \cup \{xy : x \in K,\ y \in V(G)\}$ (see [16]); we denote by S^n the composition of n disjoint copies of a prime split graph S. A graph G is *decomposable* if $G = S \circ H$ for some graphs S and H with $|S|, |H| \ge 1$, and *indecomposable* otherwise.

The *factor graph* of a split graph (S, K, I) is the loopless multigraph $\Phi = \Phi(S, K, I)$ with $V(\Phi) = I$ having one edge between u and v for each

induced P_4 of S containing both u and v [11]. The multiplicity of the edge uv is denoted by $\sigma_S(uv)$, or simply σ_{uv}. As shown in [11], the multiset $E(\Phi)$ does not depend on the bipartition, so the notation $\Phi(S)$ is unambiguous when S is active, and $\deg(S) = |\mathcal{Q}_S(P_4)| = |E(\Phi(S))|$.

Theorem 2.3 ([11])**.** *If S is an active split graph, then S is prime if and only if $\Phi(S)$ is connected.*

Proposition 2.4. *If (S, K, I) is a prime split graph, then*

$$2 \leq |I|, |K| \leq \deg(S) + 1.$$

Proof. Every member of $\mathcal{Q}_S(P_4)$ meets I in exactly two vertices, and likewise K (see [11]). If $|I| \leq 1$ or $|K| \leq 1$, then $\mathcal{Q}_S(P_4) = \varnothing$, so $\deg(S) = 0$ by Theorem 2.1 and no vertex of S is active. This gives $2 \leq |I|, |K|$.

By Theorem 2.3, $\Phi(S)$ is connected. Since $V(\Phi(S)) = I$, $|E(\Phi(S))| = \deg(S)$, and a connected multigraph of size m has order at most $m + 1$, we obtain $|I| \leq \deg(S) + 1$. Now $(\overline{S}, I, K)$ is a prime split graph with clique I and independent set K (see page 14 of [16] and page 5 of [12]). Thus, the previous argument applied to it yields $|K| \leq \deg(\overline{S}) + 1 = \deg(S) + 1$ (see [12] for the last equality). □

Lemma 2.5. *If G is a graph such that $\deg(G) = |\mathcal{Q}_G(P_4)| \leq 4$, then G is split.*

Proof. It is a well-known fact that a graph is split if and only if it contains no induced subgraphs isomorphic to C_4, $2K_2$, or C_5 (see [8]). The hypotheses imply that G contains no induced subgraphs isomorphic to C_4 or $2K_2$. If $C_5 \prec G$, then $4 \geq \deg(G) \geq \deg(C_5) = 5$. Thus, G cannot contain induced subgraphs isomorphic to C_5 either. □

A split graph S is indecomposable if and only if $\overline{S}$ is (see [16], page 14); a vertex is active in S if and only if it is active in $\overline{S}$ (see page 5 of [12]), and $\deg(\overline{S}) = \deg(S)$ (see Corollary 6.2 of [12]); and if (K, I) is the bipartition of S, then (I, K) is that of $\overline{S}$, so $\omega(\overline{S}) = \alpha(S)$ and $\alpha(\overline{S}) = \omega(S)$. Hence, to determine the prime split graphs of degree k, it suffices to find those with $\alpha(S) \leq \omega(S)$, i.e., $|I| \leq |K|$, and add the complements of those with $\alpha(S) < \omega(S)$. Those with $\alpha(S) = \omega(S)$ need not be self-complementary: for instance, R_{211} satisfies $\alpha = \omega = 3$ and $\overline{R_{211}} \not\cong R_{211}$ (see Figure 7).

The following proposition lists some basic properties of σ_{uv}.

Proposition 2.6 ([11])**.** *Let S be a split graph.*

1. $\sigma_{uv} = (d_u - \eta_{uv})(d_v - \eta_{uv})$, *where* $\eta_{uv} = |N_u \cap N_v|$;
2. $\sigma_{uv} = 0$ *and* $d_v \leq d_u$ *if and only if* $N_v \subseteq N_u$;
3. $\sigma_{uv} = 0$ *and* $d_u = d_v$ *if and only if* $N_u = N_v$;
4. *if* $\sigma_{uv} = 1$, *then* $d_u = d_v$ *and* $|N_u - N_v| = |N_v - N_u| = 1$;
5. *if* $d_u \geq d_v$ *and* $\sigma_{uv} = p$ *is prime, then* $|N_u - N_v| = p, |N_v - N_u| = 1$ *and* $d_u - d_v = p - 1$;

From now on, keep in mind that isolated and universal vertices of a graph G are not active vertices of G (see [12]).

Lemma 2.7. *Let* (S, K, I) *be an active split graph with* $I = \{a, b\}$. *Then:*

1. $|K| = d_a + d_b$ *and* $\sigma_{ab} = d_a d_b$;
2. S *is isomorphic to the prime split graph with bipartition* $([|K|], I)$ *such that* $N_a = [d_a]$ *and* $N_b = \{d_a + 1, \ldots, |K|\}$, *which we call* D_{d_b, d_a}.

Proof. 1. Since S has no universal vertices, we have $\eta_{ab} = 0$ and so $|K| = d_a + d_b$. Moreover, $\sigma_{ab} = d_a d_b$ by Proposition 2.6.

2. Straightforward. □

Lemma 2.8. *Let* (S, K, I) *be an active split graph with* $I = \{a, b, c\}$. *For* $u \in I$, *let* π_u *be the number of vertices of* K *whose only neighbor in* I *is* u, *and let* κ_u *be the number of vertices of* K *whose neighbors in* I *are exactly the two vertices of* $I - \{u\}$. *Then:*

1. $|K| = \sum_{u \in I} (\pi_u + \kappa_u)$ *and* $d_u = \pi_u + \sum_{v \in I - \{u\}} \kappa_v$ *for all* $u \in I$;
2. $\sigma_{uv} = (\pi_u + \kappa_v)(\pi_v + \kappa_u)$ *for all distinct* $u, v \in I$;
3. S *is determined, up to isomorphism, by* $(\pi_a, \pi_b, \pi_c, \kappa_a, \kappa_b, \kappa_c)$;
4. *if* $\sigma_{ac} = 0$, *then, up to exchanging* a *and* c, *we have* $\pi_a = \kappa_c = 0$, $\sigma_{ab} = \kappa_b(\pi_b + \kappa_a)$, $\sigma_{bc} = \pi_b(\pi_c + \kappa_b)$ *and* $|K| = \pi_b + \pi_c + \kappa_a + \kappa_b$.

Proof. Since S is active, every $x \in K$ is an inner vertex of an induced P_4 of S, whose ends lie in I (see [11]), so $N_x \cap I \neq \varnothing$; moreover, x is not universal, so $N_x \cap I \neq I$. Hence the six sets counted by the π_u and the κ_u partition K, which gives (1), and S is recovered, up to isomorphism, from their sizes, which gives (3). Since $|N_u - N_v| = \pi_u + \kappa_v$, (2) follows from Proposition 2.6. Finally, if $\sigma_{ac} = 0$, then by (2) we may assume, exchanging a and c if necessary, that $\pi_a + \kappa_c = 0$, and (4) follows from (1) and (2). □

Following [11], we say that a split graph (S, K, I) is *δ-homogeneous* if $\deg_S(v) = \delta$ for all $v \in I$. Recall that if S is active, then $|K| = \omega(S)$, $|I| = \alpha(S)$, and (K, I) is the unique bipartition for S. Throughout the next sections we use the following facts.

Proposition 2.9 ([12, 11])**.** *Let (S, K, I) be a split graph and let $\Phi = \Phi(S, K, I)$.*

1. *If G is a graph, then*

$$\deg((S, K, I) \circ G) = \deg(S) + \deg(G),$$

   and $S \circ G$ is active if and only if S and G are active.

2. *If $K = \bigcup_{v \in I} N_v$, $|I| \geq 2$ and Φ is simple and connected, then (S, K, I) is δ-homogeneous.*

3. *If (S, K, I) is δ-homogeneous and $\deg(S) \geq 1$, then:*

   (a) *$\sigma_{uv} = 0$ if and only if $N_u = N_v$;*

   (b) *Φ contains no three vertices u, v, w such that $\sigma_{uv} \neq 0$ and $\sigma_{uw} = 0 = \sigma_{vw}$.*

2.2 Modules and induced 4-cycles

A set $M \subseteq V(G)$ is a *module* of G if every vertex of $V(G) - M$ is adjacent either to all vertices of M or to none of them (see [9]). If M is a module of G and $W \subseteq V(G)$, then $M \cap W$ is a module of $G[W]$.

Lemma 2.10. *Let M be a module of G, let $H \prec G$ and put $W = V(H) \cap M$. If $A \subseteq M$ and $\varphi\colon W \to A$ is an isomorphism from $G[W]$ onto $G[A]$, then $G\big[(V(H) - W) \cup A\big] \approx H$.*

Proof. Extend φ by the identity on $V(H) - W$. Adjacencies inside $V(H) - W$ are unchanged and those inside W are preserved by φ. Since $V(H) - W \subseteq V(G) - M$ and M is a module, every $v \in V(H) - W$ is adjacent to all or to none of the vertices of M; hence $uv \in E(G)$ if and only if $\varphi(u)v \in E(G)$, for all $u \in W$. □

Let $A_4(G)$ be the graph on $V(G)$ in which two vertices are adjacent if and only if they lie in a common member of $\mathcal{Q}_G$, that is, if some 2-switch on G involves both of them; $A_4(G)$ is the 2-section of the A_4-structure of G introduced in [4].

Theorem 2.11 ([4]; see also [11])**.** *If $G_n \circ \cdots \circ G_1$ is the Tyshkevich decomposition of G, then $A_4(G) = \dot{\bigcup}_{i=1}^{n} A_4(G_i)$. In particular, G is indecomposable if and only if $A_4(G)$ is connected.*

Note that (2) gives $\sum_{F\in\mathcal{F}} \deg(F) \le \deg(G)$ for every $\mathcal{F} \subseteq \mathcal{Q}_G$, with equality if and only if $\mathcal{F} = \mathcal{Q}_G$. If $C \in \mathcal{Q}_G(C_4)$ and $v \in V(G) - V(C)$, define

$$\mathcal{H}_v = \big\{H \in \mathcal{Q}_G :\ V(H) \subseteq \{v\} \cup V(C) \ \text{ and } \ V(H) \neq V(C)\big\}.$$

Let $M \subseteq V(G)$. A vertex $v \in V(G) - M$ *distinguishes* M if $\varnothing \neq N_G(v) \cap M \neq M$; thus M is a module of G if and only if no vertex distinguishes it.

Lemma 2.12. *Let $C = c_0c_1c_2c_3c_0$ be an induced C_4 in G, with subscripts in $\mathbb{Z}_4$, and let $v \in V(G) - V(C)$. Then*

$$\sum_{H\in\mathcal{H}_v} \deg(H) = \begin{cases} 0, & \text{if } v \text{ does not distinguish } V(C); \\ 4, & \text{if } N_G(v) \cap V(C) = \{c_i, c_{i+2}\} \text{ for some } i \in \mathbb{Z}_4; \\ 2, & \text{otherwise.} \end{cases}$$

Proof. The 4-subsets of $\{v\} \cup V(C)$ other than $V(C)$ are exactly the sets $W_i = \{v, c_{i-1}, c_i, c_{i+1}\}$ with $i \in \mathbb{Z}_4$, so $\mathcal{H}_v = \{G[W_i] : i \in \mathbb{Z}_4\} \cap \mathcal{Q}_G$. Put $N = N_G(v) \cap V(C)$. Since $c_{i-1}c_i, c_ic_{i+1} \in E(G)$ and $c_{i-1}c_{i+1} \notin E(G)$, checking the eight possibilities for $N \cap \{c_{i-1}, c_i, c_{i+1}\}$ gives

$$G[W_i] \in \mathcal{Q}_G \iff c_i \notin N \ \text{ and } \ N \cap \{c_{i-1}, c_{i+1}\} \neq \varnothing,$$

and in that case $\deg(G[W_i]) = |N \cap \{c_{i-1}, c_{i+1}\}|$, since $G[W_i] \approx P_4$ when this number is 1 and $G[W_i] \approx C_4$ when it is 2.

If v does not distinguish $V(C)$, then $N \in \{\varnothing, V(C)\}$, no W_i satisfies the condition, and the sum is 0. Otherwise, rotating C we may assume $N \in \{\{c_0\}, \{c_0, c_1\}, \{c_0, c_2\}, \{c_0, c_1, c_2\}\}$. If $N = \{c_0, c_2\}$, then $\mathcal{H}_v = \{G[W_1], G[W_3]\}$ with both graphs isomorphic to C_4, so the sum is 4. In each of the other three cases the sum is 2: for $N = \{c_0\}$ we get $\mathcal{H}_v = \{G[W_1], G[W_3]\}$ with both graphs isomorphic to P_4; for $N = \{c_0, c_1\}$ we get $\mathcal{H}_v = \{G[W_2], G[W_3]\}$, again two copies of P_4; and for $N = \{c_0, c_1, c_2\}$ we get $\mathcal{H}_v = \{G[W_3]\}$ with $G[W_3] \approx C_4$. □

Theorem 2.13. *Let $C \prec G$ with $C \approx C_4$. Then*

$$\deg(G) \ \ge \ 2 + \sum_{v\in V(G)-V(C)} \ \sum_{H\in\mathcal{H}_v} \deg(H), \tag{3}$$

with equality if and only if $\mathcal{Q}_G = \{C\} \cup \bigcup_{v \in V(G)-V(C)} \mathcal{H}_v$. *Consequently, at most* $\lfloor (\deg(G) - 2)/2 \rfloor$ *vertices distinguish* $V(C)$. *In particular,* $V(C)$ *is a module of* G *if* $\deg(G) \leq 3$*; and if* $\deg(G) \leq 5$*, then at most one vertex* v *distinguishes* $V(C)$*, and* $N_G(v) \cap V(C)$ *is not a pair of opposite vertices of* C.

Proof. Every member of $\mathcal{H}_v$ contains v and exactly three vertices of C; hence the families $\mathcal{H}_v$ are pairwise disjoint and none of them contains C. Since $\deg(C) = 2$, (3) and its equality case follow from (2).

By Lemma 2.12, a vertex contributes 0 to the sum in (3) if it does not distinguish $V(C)$, and at least 2 if it does. If t vertices distinguish $V(C)$, then (3) gives $\deg(G) \geq 2 + 2t$, so $t \leq \lfloor (\deg(G) - 2)/2 \rfloor$. For $\deg(G) \leq 3$ this yields $t = 0$, that is, $V(C)$ is a module; for $\deg(G) \leq 5$ it yields $t \leq 1$. Finally, if $\deg(G) \leq 5$ and a vertex v had $N_G(v) \cap V(C)$ equal to a pair of opposite vertices of C, then its contribution would be 4 by Lemma 2.12, whence $\deg(G) \geq 2 + 4 = 6$, a contradiction. □

Lemma 2.14. *Let* G *be a prime graph and let* $C \prec G$ *with* $C \approx C_4$. *If* $V(C)$ *is a module of* G *and* $G \neq C$*, then* $\deg(G) \geq 6$.

Proof. Put $M = V(C)$. Since $G \neq C$, we have $V(G) - M \neq \varnothing$, and since $A_4(G)$ is connected by Theorem 2.11, some $H \in \mathcal{Q}_G$ meets both M and $V(G) - M$. Put $W = V(H) \cap M$, so that $1 \leq |W| \leq 3$. If $|W| = 3$, the vertex of $V(H) - W$ is adjacent to all or to none of the vertices of W, which is impossible because the vertices of P_4, C_4 and $2K_2$ have degree 1 or 2. If $|W| = 2$, say $W = \{w_1, w_2\}$, then since M is a module every vertex of $V(H) - W$ is adjacent to both or to neither of w_1, w_2; as no two vertices of P_4 have this property, $\deg(H) = 2$. By Lemma 2.10, each $A \subseteq M$ with $G[A] \approx G[W]$ yields a member $G[(V(H) - W) \cup A]$ of $\mathcal{Q}_G$ isomorphic to H and distinct from C; since C_4 has four pairs of vertices inducing an edge and two inducing a non-edge, there are at least two such A, whence $\deg(G) \geq 2 + 2 \cdot 2 = 6$ by (2). If $|W| = 1$, Lemma 2.10 applied to the four singletons $A \subseteq M$ yields four members of $\mathcal{Q}_G$ distinct from C, so $\deg(G) \geq 2 + 4 = 6$. □

3 Active graphs of degree 1 and 2

An immediate consequence of Theorem 2.2 is that P_4 is the only active graph of degree 1. Then, by Proposition 2.9, P_4^2 is the only decomposable active graph of degree 2. Now, consider a prime graph G such that $\deg(G) = |\mathcal{Q}_G(P_4)| = 2$. By Lemma 2.5, G is split.

Then $\Phi(G)$ is connected, by Theorem 2.3. This means that $\Phi(G)$ is isomorphic to K_2 or P_3, ignoring multiplicities. If (K, I) is the bipartition for G, we have $(|K|, |I|) = (\omega, \alpha)$, where $\omega = \omega(G)$ and $\alpha = \alpha(G)$. Assume $\alpha \leq \omega$. As $5 \leq \alpha + \omega = |G|$, it follows that $\omega \neq 2$. Thus, $(\omega, \alpha) \in \{(3, 2), (3, 3)\}$ by Proposition 2.4. If $\alpha = 2$, then $G \approx D_{2,1}$ by Lemma 2.7 (see Figure 1). If $\alpha = 3$, then $\Phi = abc$, and Lemma 2.8 gives $\kappa_b(\pi_b + \kappa_a) = 1 = \pi_b(\pi_c + \kappa_b)$. Hence $\pi_b = \kappa_b = 1$ and $\kappa_a = \pi_c = 0$, so $\omega = 2$, which is absurd.

So, if G is a prime split graph of degree 2, then $G \approx D_{2,1}$ or $\overline{D_{2,1}}$. If G is a prime non-split graph of degree 2, it is clear that $|\mathcal{Q}_G| = 1$. Moreover, $|G| = 4$, since otherwise $|\mathcal{Q}_G| \geq 2$. Then, $G \approx C_4$ or $2K_2$.

Theorem 3.1. *Let G be an active graph.*

1. *If $\deg(G) = 1$, then $G \approx P_4$.*
2. *If $\deg(G) = 2$ and G is decomposable, then $G \approx P_4^2$.*
3. *If $\deg(G) = 2$ and G is indecomposable, then G is isomorphic to one of these graphs: $D_{2,1}$, $\overline{D_{2,1}}$, C_4, $2K_2$.*

All graphs mentioned here are represented in Figure 1.

Proof. It follows from the previous discussion. □

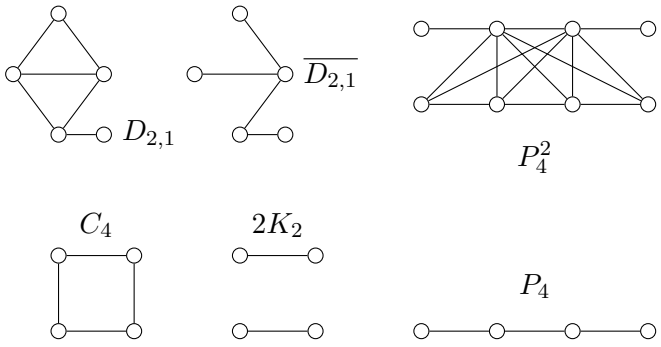


Figure 1: The 6 active graphs of degree 1 or 2.

4 Active graphs of degree 3

Consider an active graph G such that $\deg(G) = |\mathcal{Q}_G(P_4)| = 3$. Thanks to Lemma 2.5, we know that G is split. Suppose G is decomposable. Since G

is active, it follows from Proposition 2.9 that each factor of G is an active split graph of degree 1 or 2. Consequently,

$$G \in \{P_4 \circ D_{2,1}, D_{2,1} \circ P_4, P_4 \circ \overline{D_{2,1}}, \overline{D_{2,1}} \circ P_4, P_4^3\},$$

by Theorem 3.1.

If instead (G, K, I) is indecomposable, assume $\alpha = |I| \leq |K| = \omega$. Clearly, $\omega \geq 3$, since otherwise we would have $G \approx P_4$. Thus, $2 \leq \alpha \leq 4$ and $\omega \in \{3, 4\}$, by Proposition 2.4. Applying Theorem 2.3, we have

$$\Phi(G) \in \{K_2, K_3, P_3, K_{3,1}, P_4\},$$

ignoring multiplicities if any (see Figure 2). If $\alpha = 2$, then $G \approx D_{3,1}$ by

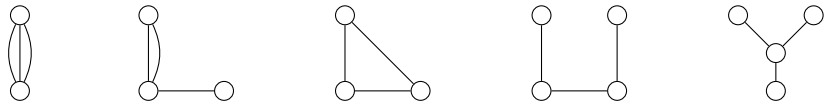


Figure 2: The 5 connected multigraphs of size 3.

Lemma 2.7 (see Figure 3). If $\Phi = \Phi(G) \in \{K_3, K_{3,1}, P_4\}$, then G is δ-homogeneous, by Proposition 2.9. The same proposition therefore prohibits $\Phi = P_4$. Suppose $\Phi = K_{3,1}$. Thanks to Proposition 2.9, we know that $N_1 = N_2 = N_3$. On the other hand, $0 = |\bigcap_{i=1}^4 N_i| = \eta_{14}$, because G has no universal vertices. This means that $\delta = 1$, by Proposition 2.6. But then $\omega = 2$, contradicting $\alpha \leq \omega$. Hence, $\alpha \neq 4$.

If $\alpha = 3$, then $\Phi = abca$, or $\Phi = abc$ with $\{\sigma_{ab}, \sigma_{bc}\} = \{1, 2\}$. If $\Phi = abca$ then the six factors in Lemma 2.8(2) are equal to 1, so $\pi_a = \pi_b = \pi_c$, $\kappa_a = \kappa_b = \kappa_c$ and $\pi_a + \kappa_a = 1$. This yields two graphs, complementary to each other; we call T_{111} the one with $\pi_a = 1$ (the subscript records the degrees of the I-vertices). If $\Phi = abc$, then, exchanging a and c if necessary, Lemma 2.8(4) gives $\sigma_{ab} = \kappa_b(\pi_b + \kappa_a)$ and $\sigma_{bc} = \pi_b(\pi_c + \kappa_b)$, so $\pi_b = \kappa_b = 1$. If $\sigma_{ab} = 2$, then $\kappa_a = 1$ and $\pi_c = 0$, and we call the resulting graph T_{221}; if $\sigma_{ab} = 1$, then $\kappa_a = 0$ and $\pi_c = 1$, which yields $\overline{T_{221}}$. In all cases $\omega = 3$. These graphs are shown in Figure 3.

So far, we have classified active graphs G of degree 3 such that $\mathcal{Q}_G = \mathcal{Q}_G(P_4)$. To complete the classification of all active graphs of degree 3, it remains to analyze the following case:

$$|\mathcal{Q}_G(C_4)| + |\mathcal{Q}_G(2K_2)| = 1 = |\mathcal{Q}_G(P_4)|. \tag{4}$$

Recall that, by definition of $\circ$, if $G = X \circ Y$, then X is split. So, if G is decomposable and satisfies (4), then $G \in \{P_4 \circ C_4, P_4 \circ 2K_2\}$, by Theorem 3.1

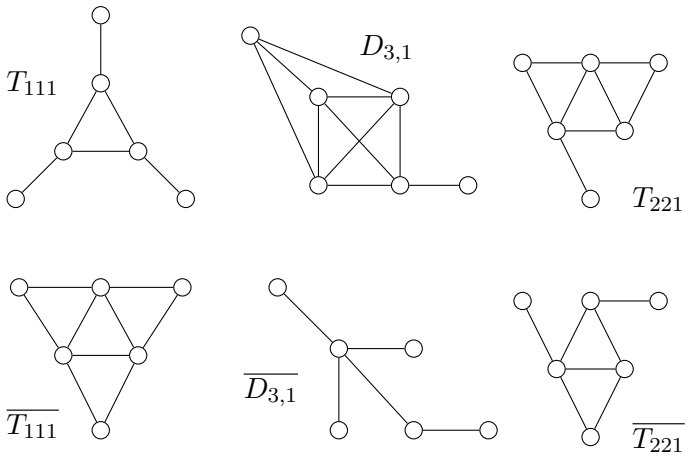


Figure 3: The 6 prime graphs of degree 3.

and Proposition 2.9. In the following lemma, we prove that there are no prime graphs satisfying (4).

Lemma 4.1. *There exists no prime graph satisfying* (4)*.*

Proof. Assume by contradiction that G is prime and satisfies (4). A 4-subset of $V(G)$ induces P_4, C_4 or $2K_2$ in G if and only if it induces P_4, $2K_2$ or C_4 in $\overline{G}$, respectively, and G is prime if and only if $\overline{G}$ is; hence we may assume $\mathcal{Q}_G = \{P, C\}$ with $P \approx P_4$ and $C \approx C_4$. By (2), $\deg(G) = 3$, so $V(C)$ is a module of G by Theorem 2.13. Moreover $G \neq C$, because $\deg(C) = 2 \neq \deg(G)$. Therefore $\deg(G) \geq 6$ by Lemma 2.14, a contradiction. □

Theorem 4.2. *Let G be an active graph of degree 3.*

1. *If G is decomposable, then $G \in$*

$$\{P_4 \circ D_{2,1}, D_{2,1} \circ P_4, P_4 \circ \overline{D_{2,1}}, \overline{D_{2,1}} \circ P_4, P_4^3, P_4 \circ C_4, P_4 \circ 2K_2\},$$

2. *If G is indecomposable, then*

$$G \in \{T_{111}, \overline{T_{111}}, D_{3,1}, \overline{D_{3,1}}, T_{221}, \overline{T_{221}}\},$$

all of which are split (see Figure 3).

Proof. It follows from the previous discussion and Lemma 4.1. □

5 Prime split graphs of degree 4

We now classify the prime split graphs of degree 4. Consider a prime graph G such that $\deg(G) = |\mathcal{Q}_G(P_4)| = 4$. Thanks to Lemma 2.5, we know that G is split. Assume G has bipartition (K, I) and $\alpha = |I| \leq |K| = \omega$. Clearly, $\omega \geq 3$, since otherwise we would have $G \approx P_4$. Thus, $2 \leq \alpha \leq 5$ and $3 \leq \omega \leq 5$, by Proposition 2.4. Applying Theorem 2.3, we have that $\Phi = \Phi(G)$ is isomorphic to one of the 12 multigraphs in Figure 4. If $\alpha = 2$, then

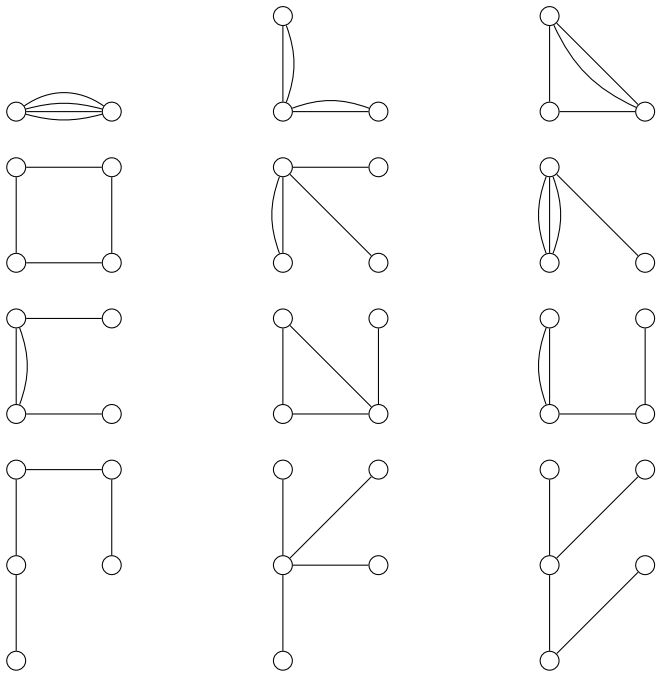


Figure 4: The 12 connected multigraphs of size 4.

$G \approx D_{4,1}$ or $D_{2,2}$ by Lemma 2.7 (see Figure 5). Thanks to Proposition 2.9, we can immediately rule out that Φ is the K_3 with a pendant vertex, P_5, or $\overline{D_{2,1}}$. Applying Theorem 4.1 of [13], we can also exclude that $\Phi = v_1v_2v_3v_4$ with $\sigma_{23} = 1$. If $\Phi = abca$ with $\sigma_{ac} = 2$, then by Proposition 2.6 we would have $d_a = d_b = d_c = d_a \pm 1$, which is absurd.

If $\Phi \approx K_{4,1}$, let v_5 be the vertex of degree 4 in $K_{4,1}$ and v_i its leaves, $i \in [4]$. Applying Proposition 2.9, we have that the $N_i = N_j$ for all $i, j \in [4]$, so $K = N_1 \cup N_5$. On the other hand, since G has no universal vertices, we have $\varnothing = \bigcap_{j=1}^{5} N_j = N_1 \cap N_5$, and consequently, $d_1 = 1 = d_5$, by Proposition 2.6. Then, $5 = \alpha \leq \omega = 2$, an absurdity.

If $\Phi = v_1v_2v_3v_4v_1$, then $N_1 = N_3$ and $N_2 = N_4$, by Proposition 2.9, so $K = N_1 \cup N_2$. Since G has no universal vertices, we have $\varnothing = \bigcap_{i=1}^{4} N_i = N_1 \cap N_2$, and consequently, $d_1 = 1 = d_2$, by Proposition 2.6. Then, $4 = \alpha \leq \omega = d_1 + d_2 = 2$, which is impossible.

If $\Phi = v_1v_2v_3v_4$ with $\sigma_{23} = 2$, then we can assume without loss of generality that $d_2 \leq d_3$. Then, $d_1 = d_2$, $d_3 = d_1 + 1 = d_4$, $N_2 \subseteq N_4$, and $N_1 \subseteq N_3, N_4$, by Proposition 2.6. Thus, $K = N_3 \cup N_4$. Since G has no universal vertices, we have $\varnothing = \bigcap_{i=1}^{4} N_i = N_1 \cap N_2$, and consequently, $d_1 = 1$, by Proposition 2.6. Thus, $d_3 = 2$ and $\eta_{34} = 1$, again by Proposition 2.6. Finally, we obtain $4 = \alpha \leq \omega = d_3 + d_4 - \eta_{34} = 3$, an absurdity.

If $E(\Phi) = \{2v_1v_2, v_2v_3, v_2v_4\}$ (a $K_{3,1}$ with a double-edge), then $d_3 = d_2 = d_4$, $|d_2 - d_1| = 1$, and $N_3 = N_4$, by Proposition 2.6. If $d_2 = d_1 + 1$, then $N_1 \subseteq N_3$, so $K = N_2 \cup N_3$. Since G has no universal vertices, we have $\varnothing = \bigcap_{i=1}^{4} N_i = N_1 \cap N_2$, and consequently, $d_1 = 1$, by Proposition 2.6. Thus, $\eta_{23} = 1$, again by Proposition 2.6. Finally, we obtain $4 = \alpha \leq \omega = d_2 + d_3 - \eta_{23} = 3$, which is impossible. The case $d_1 = d_2 + 1$ is reduced to an absurdity with similar arguments.

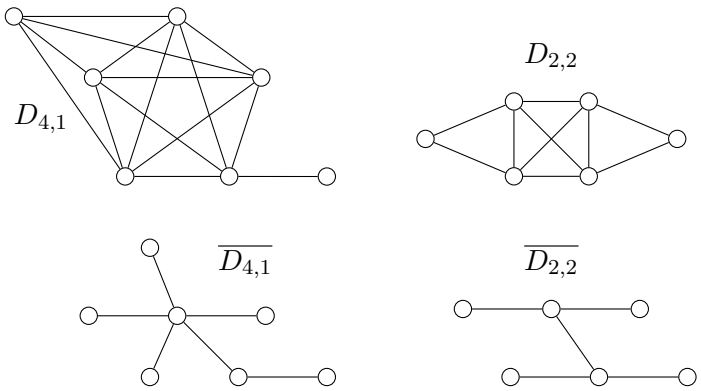


Figure 5: The prime split graphs $D_{4,1}, D_{2,2}$ and their complements.

If $\Phi = abc$ ignoring multiplicities, then, exchanging a and c if necessary, Lemma 2.8 gives $\sigma_{ab} = \kappa_b(\pi_b + \kappa_a)$ and $\sigma_{bc} = \pi_b(\pi_c + \kappa_b)$, with $\sigma_{ab} + \sigma_{bc} = 4$ and $\pi_b, \kappa_b \geq 1$. A direct computation yields exactly the five graphs of the following table, where only the parameters of Lemma 2.8 not forced to vanish are listed; as in Section 4, the subscript of each name records the degrees of the I-vertices. In all cases $\omega \geq 3 = \alpha$.

$(\sigma_{ab}, \sigma_{bc})$	$(\pi_b, \pi_c, \kappa_a, \kappa_b)$	ω	G
$(1,3)$	$(1,2,0,1)$	4	R_{311}
$(3,1)$	$(1,0,2,1)$	4	R_{331}
$(2,2)$	$(1,1,1,1)$	4	R_{321}
$(2,2)$	$(2,0,0,1)$	3	R_{211}
$(2,2)$	$(1,0,0,2)$	3	$\overline{R_{211}}$

These graphs, together with $\overline{R_{311}}$, $\overline{R_{331}}$ and $\overline{R_{321}}$, are shown in Figures 6 and 7.

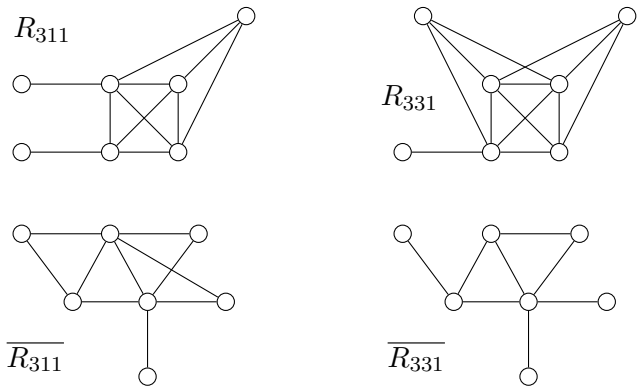


Figure 6: The prime split graphs R_{311}, R_{331} and their complements.

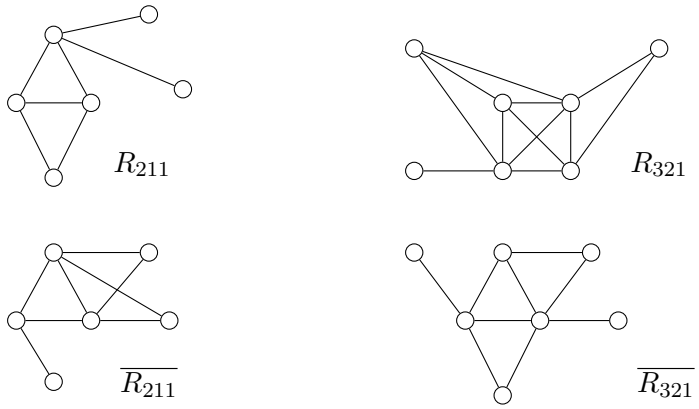


Figure 7: The prime split graphs R_{211}, R_{321} and their complements.

Theorem 5.1. *The 12 graphs in Figures 5 to 7 are all the prime split graphs of degree 4.*

Proof. It follows from the previous discussion. □

6 Prime non-split graphs of degree 4

For $j \in [3]$, let B_j be the graph with vertex set $\{v, c_0, c_1, c_2, c_3\}$ in which $c_0c_1c_2c_3c_0$ is an induced C_4 and $N_{B_j}(v) = \{c_0, \dots, c_{j-1}\}$. These six graphs are displayed in Figure 8. In this section we show that they are all the prime non-split graphs of degree 4.

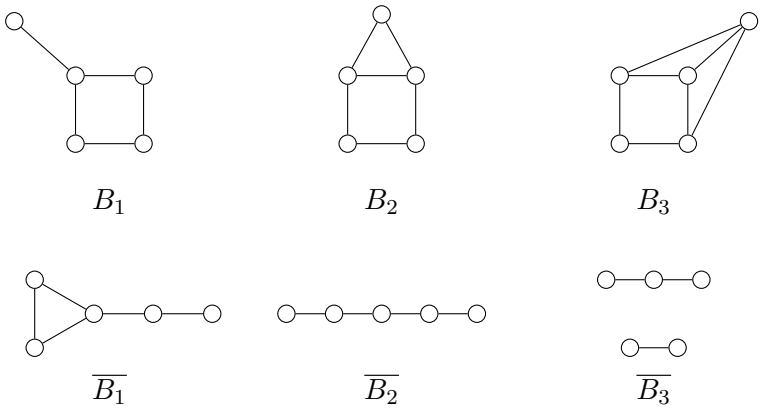


Figure 8: The six non-split prime graphs of degree 4.

Theorem 6.1. *The graphs B_1, B_2, B_3 and their complements (see Figure 8) are all the non-split prime graphs of degree 4.*

Proof. Fix $j \in [3]$ and write $C = B_j[c_0, c_1, c_2, c_3]$. As $|B_j| = 5$, every member of $\mathcal{Q}_{B_j}$ either equals C or contains v; hence $\mathcal{Q}_{B_j} = \{C\} \cup \mathcal{H}_v$, so $\deg(B_j) = 2 + 2 = 4$ by Lemma 2.12 and (2), and B_j is active. If $B_j = S \circ H$ with $V(S), V(H) \neq \varnothing$, then S and H are active by Proposition 2.9, so $|S|, |H| \geq 4$ by Theorem 2.2 and $|B_j| \geq 8$, which is absurd; hence B_j is prime. Moreover B_j is not split, because $C \approx C_4$. Since complementation preserves the degree, primality and non-splitness, the same holds for $\overline{B_1}, \overline{B_2}, \overline{B_3}$. Finally, these six graphs are clearly pairwise non-isomorphic.

Conversely, let G be a prime graph with $\deg(G) = 4$ which is not split. By Lemma 2.5 we have $\deg(G) \neq |\mathcal{Q}_G(P_4)|$, so $\mathcal{Q}_G(C_4) \cup \mathcal{Q}_G(2K_2) \neq \varnothing$ by Theorem 2.1. Complementation preserves the degree, primality and non-splitness, and interchanges induced C_4's with induced $2K_2$'s; as the six graphs of the statement form three complementary pairs, we may assume $\mathcal{Q}_G(C_4) \neq \varnothing$

and it suffices to prove that $G \approx B_j$ for some $j \in [3]$. Fix an induced $C = c_0c_1c_2c_3c_0 \approx C_4$ in G and put $M = V(C)$.

By Theorem 2.13, at most one vertex $v_0 \in V(G) - M$ distinguishes M, and $N_{v_0} \cap M$ is not a pair of opposite vertices of C; by Lemma 2.12, $\sum_{H \in \mathcal{H}_{v_0}} \deg(H) = 2$.

Suppose first that no such v_0 exists. Then no vertex of $V(G) - M$ distinguishes M, so M is a module of G; moreover $G \neq C$ because $\deg(C) = 2$. This contradicts Lemma 2.14. Hence v_0 exists, and (3) becomes an equality, so $\mathcal{Q}_G = \{C\} \cup \mathcal{H}_{v_0}$. In particular every member of $\mathcal{Q}_G$ has all its vertices in $\{v_0\} \cup M$, and since G is active we get $V(G) = \{v_0\} \cup M$. Finally, after a rotation of C we may assume $N_{v_0} \cap M = \{c_0, \ldots, c_{j-1}\}$ for some $j \in [3]$, because $N_{v_0} \cap M$ is neither empty, nor M, nor a pair of opposite vertices of C. Therefore $G \approx B_j$. □

Corollary 6.2. *There are exactly 18 prime graphs of degree 4: the 12 split graphs of Figures 5 to 7, together with the 6 non-split graphs of Figure 8.*

7 Consequences for realization graphs

We now apply the previous classifications to unigraphs and realization graphs; if X has degree sequence d, we also write $\mathcal{G}(X)$ for $\mathcal{G}(d)$. Following [16], a degree sequence is *unigraphical* if all its realizations are isomorphic, and a *unigraph* is a realization of a unigraphical sequence; following [5], a graph is a *hereditary unigraph* if all its induced subgraphs are unigraphs, and it is *$\mathcal{F}$-free* if none of its induced subgraphs lies in the family $\mathcal{F}$. Let

$$\mathcal{F} = \big\{H, \overline{H} : \ H \in \{P_5,\ K_2 \dot\cup K_3,\ B_1,\ 2P_3,\ K_2 \dot\cup P_4,\ K_2 \dot\cup C_4,\ \overline{T_{221}},\ R_{211}\}\big\}.$$

In [5] $\overline{T_{221}}$ and R_{211} are called R and S (see Figure 6 of [5]).

Theorem 7.1 ([5], Theorem 4.2)**.** *A graph is a hereditary unigraph if and only if it is $\mathcal{F}$-free.*

If $H \prec G$, then $\mathcal{Q}_H \subseteq \mathcal{Q}_G$, so (2) gives

$$H \prec G \implies \deg(H) \leq \deg(G). \tag{5}$$

Theorem 7.2. *Let $\deg(G) \leq 3$. Then G is a unigraph if and only if it is a hereditary unigraph, if and only if it contains neither T_{221} nor $\overline{T_{221}}$ as an induced subgraph; and otherwise some 2-switch τ on G satisfies $\deg(\tau(G)) \geq 4$. In particular, every graph of degree at most 2 is a hereditary unigraph.*

Proof. As $\deg(\overline{H}) = \deg(H)$ for every H [12], the members of $\mathcal{F}$ other than $T_{221}, \overline{T_{221}}$ have degree at least 4. Hence, if $T_{221}, \overline{T_{221}} \not\prec G$, then G is $\mathcal{F}$-free by (5), so it is a hereditary unigraph by Theorem 7.1, and in particular a unigraph; the same argument proves the last assertion, because (5) forbids $T_{221}, \overline{T_{221}}$ inside a graph of degree at most 2.

Assume now $T_{221} \prec G$ or $\overline{T_{221}} \prec G$. A 2-switch on $\overline{G}$ is a 2-switch on G on the same 4 vertices, the resulting graphs being complementary, so we may assume $T_{221} \prec G$. It is easy to check that there is a 2-switch τ on T_{221} such that $\tau(T_{221}) \approx \overline{R_{211}}$. Since $\tau(T_{221}) \prec \tau(G)$, we have $\deg(\tau(G)) \geq 4$ by (5), and since $\tau(G)$ and G share their degree sequence, G is not a unigraph. □

Corollary 7.3. *Let X realize d, with $\deg(X) = k \leq 3$. Then $\mathcal{G}(d)$ is regular if and only if $T_{221}, \overline{T_{221}} \not\prec X$, and in that case d is unigraphical, $\mathcal{G}(d)$ is vertex-transitive and k-regular, and $\tau(X) \approx X$ for every 2-switch τ on X.*

Proof. By Theorem 7.2, d is unigraphical. Let Γ be the group of permutations π of $V(X)$ with $d_{\pi(v)} = d_v$ for all v. Relabeling by $\pi \in \Gamma$ sends realizations of d to realizations of d and preserves the relation of differing by a 2-switch, so Γ acts on $\mathcal{G}(d)$ by automorphisms; and any isomorphism between realizations of d preserves degrees, hence lies in Γ. So Γ is transitive, $\mathcal{G}(d)$ is vertex-transitive, and it is k-regular because X has degree k in it. The last assertion follows from Theorem 7.2. □

Thus $\mathcal{G}(T_{221}) \approx K_{1,2,2}$: four of its vertices are isomorphic to T_{221} and have degree 3, while the fifth is isomorphic to $\overline{R_{211}}$ and has degree 4; so T_{221} and $\overline{R_{211}}$ are the two realizations of $d(T_{221})$ up to isomorphism. This is the first instance of a phenomenon occurring in every degree $k \geq 3$.

Proposition 7.4. *For every $k \geq 3$ there are a prime graph G_k with $\deg(G_k) = k$ and a 2-switch τ on G_k with $\deg(\tau(G_k)) = 2k - 2$; in particular, $\mathcal{G}(G_k)$ is not regular.*

Proof. Let G_k be the split graph with clique $K = \{x, y_1, \ldots, y_{k-1}\}$, independent set $I = \{a, b, c\}$, and $N_a = \{x\}$, $N_b = \{y_1, \ldots, y_{k-1}\}$, $N_c = \{y_1\}$. By Proposition 2.6, $\sigma_{ab} = k - 1$, $\sigma_{ac} = 1$ and $\sigma_{bc} = 0$, so $\deg(G_k) = k$ and $\Phi(G_k) = cab$ ignoring multiplicities. Every vertex of $K \cup \{a, b\}$ lies in an induced P_4 axy_ib, and c in the induced P_4 axy_1c; hence G_k is active, and prime by Theorem 2.3. Let τ replace ax, by_1 with ay_1, bx. In $\tau(G_k)$ we have $N_a = N_c = \{y_1\}$ and $N_b = \{x, y_2, \ldots, y_{k-1}\}$, so $\sigma_{ab} = k - 1 = \sigma_{cb}$ and $\sigma_{ac} = 0$, whence $\deg(\tau(G_k)) = 2k - 2 \neq k$. □

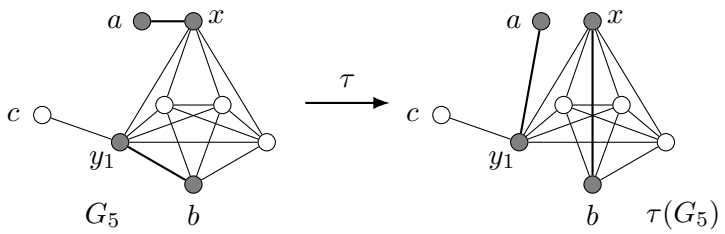


Figure 9: The prime split graph G_5 and its image under the 2-switch τ, which replaces ax, by_1 with ay_1, bx.

Note that $G_3 \approx \overline{T_{221}}$ and $G_4 \approx R_{311}$ (see Figures 3 and 6). In Figure 9, G_5 and $\tau(G_5)$ are shown.

We now turn to the realization graphs themselves. The first result below reduces their study to prime graphs; in [2] it is stated for the canonical decomposition of d, whose components are the degree sequences of $G_n, \ldots, G_1$.

Theorem 7.5 ([2], Theorem 6)**.** *If $G_n \circ \cdots \circ G_1$ is the Tyshkevich decomposition of X, then $\mathcal{G}(X) \approx \mathcal{G}(G_n) \,\square\, \cdots \,\square\, \mathcal{G}(G_1)$.*

Theorem 7.6 ([3], page 2)**.** $\mathcal{G}(d) \approx \mathcal{G}(\overline{d})$.

The classification obtained in the previous sections determines the realization graph of every prime graph of degree at most 4; by Theorem 7.6, only one sequence in each complementary pair has to be treated.

Theorem 7.7. *Let X be a prime graph with* $\deg(X) \leq 4$. *Then $\mathcal{G}(X)$ is isomorphic to one of the ten graphs of Figure 10, according to the following table.*

$\mathcal{G}(X)$	X
K_2	P_4
K_3	C_4, $2K_2$, $D_{2,1}$, $\overline{D_{2,1}}$
K_4	$D_{3,1}$, $\overline{D_{3,1}}$
$K_{3,3}$	T_{111}, $\overline{T_{111}}$
$K_{1,2,2}$	T_{221}, $\overline{T_{221}}$, R_{211}, $\overline{R_{211}}$
K_5	$D_{4,1}$, $\overline{D_{4,1}}$
$K_{2,2,2}$	$D_{2,2}$, $\overline{D_{2,2}}$, B_1, $\overline{B_1}$, B_3, $\overline{B_3}$
$K_{1,3,3}$	B_2, $\overline{B_2}$
$\mathcal{G}_1$	R_{311}, $\overline{R_{311}}$, R_{331}, $\overline{R_{331}}$
$\mathcal{G}_2$	R_{321}, $\overline{R_{321}}$

Proof. By Theorems 3.1, 4.2, 5.1 and 6.1 there are exactly 29 prime graphs of degree at most 4, and they realize 26 distinct degree sequences: the only coincidences are $d(T_{221}) = d(\overline{R_{211}})$, $d(\overline{T_{221}}) = d(R_{211})$ and $d(B_1) = d(\overline{B_1})$. For each of these sequences, listing the realizations of d and the 2-switches between them is a finite computation, which yields the table. □

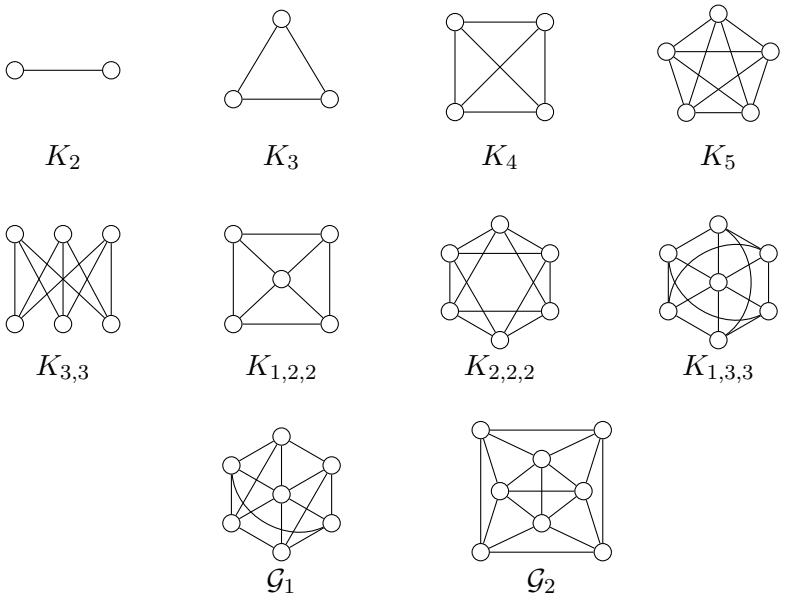


Figure 10: The ten realization graphs of Theorem 7.7.

The four graphs whose realization graph is $K_{1,2,2}$ are precisely the four members of $\mathcal{F}$ drawn in Fig. 6 of [5], and T_{111} is the 3-net, so $\mathcal{G}(T_{111}) \approx K_{3,3}$ is the transposition graph T_3, in accordance with Example 3 of [2]. In degree 4 the converse of Corollary 7.3 fails: $\mathcal{G}(B_1) \approx K_{2,2,2}$ is regular, yet $d(B_1) = d(\overline{B_1})$ with $B_1 \not\approx \overline{B_1}$.

Corollary 7.8. *If $\mathcal{G}(d)$ has a vertex X of degree $k \in [4]$, then $\mathcal{G}(d)$ is the Cartesian product of the realization graphs of the prime factors of X, each of which is one of the ten graphs of Figure 10. In particular $\mathcal{G}(d) \approx K_2$ if $k = 1$, and $\mathcal{G}(d) \approx K_3$ or C_4 if $k = 2$.*

Proof. Let $G_n \circ \cdots \circ G_1$ be the Tyshkevich decomposition of X. If $|G_i| = 1$, then $\mathcal{G}(G_i) \approx K_1$, a neutral factor for $\square$; otherwise $A_4(G_i)$ is connected by Theorem 2.11, so every vertex of G_i lies in a member of $\mathcal{Q}_{G_i}$ and G_i is prime. The prime factors $X_1, \ldots, X_m$ of X thus satisfy $\sum_i \deg(X_i) = k \leq 4$ by

Proposition 2.9, and the first assertion follows from Theorems 7.5 and 7.7. If $k = 1$, then $m = 1$ and $X_1 \approx P_4$; if $k = 2$, then either $m = 1$ and $\mathcal{G}(X_1) \approx K_3$, or $m = 2$ and $\mathcal{G}(d) \approx K_2 \,\square\, K_2 \approx C_4$. □

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